

Program : Diploma in Wood and Paper Technology	
Course Code : 2129	Course Title: Botany Lab
Semester : 2	Credits: No Credit
Course Category: Engineering Science	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To impart knowledge on plant morphology, taxonomy and anatomy.
- To provide students with skills necessary to section and stain fresh plant material in preparation for study of plant anatomy.
- To train students in the proper use of the compound light microscope and to give them experience in interpreting images that they see through the microscope in terms of how plant structure is related to function.

Course Prerequisites:

Topic	Course code	Course name	Semester
Morphology, Taxonomy		Secondary School	

Course Outcomes:

On Completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive level
CO1	Interpret Plant diversity and morphology of root, stem and leaves	9	Applying
CO2	Analyse reproductive morphology and morphology of fruits	9	Applying
CO3	Identify angiosperms Taxonomy	12	Applying
CO4	Interpret Plant anatomy	12	Applying
	Lab Exam	3	

CO-POMapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1				3			
CO2				3			
CO3				3			
CO4				3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Interpret Plant diversity and morphology of root, stem and leaves.		
M1.01	Perform microscopic observation of algae – Spirogyra, Fungi-Rhizopus, Bryophyta-Moss, Pteridophyta Nephrolepis	3	Applying
M1.02	Conduct morphological study of root, stem and leaves	6	Applying
CO2	Analyse reproductive morphology and morphology of fruits		
M2.01	Examine the different Inflorescence, flowers and fruits	9	Applying
	Lab Exam – I	1.5	
CO3	Identify angiosperms Taxonomy		
M3.01	Conduct study of vegetative and floral characters of any one number each of the families (Annonaceae, Euphorbiaceae, Verbanaceae, Rutaceae, Fabeaceae, Myrtaceaeand, Graminea)	12	Applying
CO4	Interpret Plant anatomy		
M4.01	Demonstrate Plant cell-onion peel, Tissues, C.S.and L.S, Eupatorium/Cucurbit Stem.	2	Applying
M4.02	Examine Primary structure of Dicot and monocot Stem, dicot and Monocot root of various wood samples	3	Applying
M4.03	Analyse primary structure of Dicot leaf –Ixora, Monocot leaf-Grass	2	Applying

M4.04	Identify Secondary thickening in Dicot stem- Vernonia/Eupatorium	3	Applying
M4.05	Demonstrate Heart Wood and Sap Wood; Tyloses; Hard Wood and Soft Wood; Growth Rings and Dendrochronology.	2	Understanding
	Lab Exam– II	1.5	

Text / Reference:

T/R	Book Title/Author
T1	Singh, Pande, Jain, Text book of Botany
R1	A.M.Bendre, Practical botany, Rastogi Publications
R2	Dr.Rekha Arya, Practical Botany, Hindhisahithyasadan

Online Resources:

Sl. No	Website Link
1	https://www.youtube.com/watch?v=PwrADWcoyTs&list=PLLy_2iUCG87CTiGgwV-TU0zWDBOGm_911&ab_channel=IITRoorkeeJuly2018
2	https://cdn1.byjus.com/wp-content/uploads/2019/04/Tamilnadu-Board-Class-12-Botany-Practical.pdf
3	https://ncert.nic.in/science-laboratory-manual.php